

MODEL EXIT EXAM 2026

Official Verified Answer Key & Comprehensive Explanations Document

Total Questions: 52 Extracted Entries | Prepared for Academic Assessment Support

QUICK ANSWER REFERENCE MATRIX

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	D	2	A	3	C	4	C	5	C	6	A
7	B	8	C	9	B	10	C	11	B	12	C
13	B	14	C	15	B	16	B	17	B	18	D
19	B	20	A	21	A	22	B	23	A	24	B
25	A	26	B	27	B	28	E	29	C	30	C
31	E	32	C	33	C	34	B	35	E	36	C
37	D	38	C	39	B	40	C	41	C	42	B
43	C	44	C	45	B	46	C	47	B	48	B
49	C	50	B	51	C	52	B	53	B	54	C
55	B	56	C	57	A	58	C	59	B	60	D
61	B	62	B	63	C	64	B	65	B	66	B
67	B	68	B	69	C	70	B	71	B	72	C
73	B	103	A	105	D	106	B				

DETAILED QUESTIONS & SOLUTIONS

Q1. In a CIDR block of 192.168.100.0/23, what is the broadcast address?

- A. 192.168.102.0
- B. 192.168.100.1
- C. 192.168.100.255

✓ **D. 192.168.101.255**

Explanation: A /23 block covers 192.168.100.0–192.168.101.255 (512 addresses). The broadcast is the last address: 192.168.101.255. Options a/b/c are wrong because they fall outside or at incorrect positions within this range.

Q2. Which of the following describes the possible alternative ways in which files on a disk can be accessed?

✓ **A. serial or random**

- B. object or serial
- C. text or binary
- D. binary or random
- E. random or binary

Explanation: Files on disk are accessed either serially (sequentially, one record after another) or randomly (directly jumping to any record). The other options mix file type concepts (text/binary) with access methods, which are unrelated.

Q3. In Agile development, what is a sprint?

- A. A testing phase
- B. A debugging session

✓ **C. A time-boxed development cycle**

- D. A rapid design process

Explanation: A sprint is the core unit of Agile/Scrum: a fixed-length (usually 2–4 week) iteration during which a defined set of work is completed. It is not limited to testing, debugging, or design alone.

Q4. Decision trees classify data points by recursively splitting the data space based on _____ features.

- A. numerical
- B. most relevant

✓ **C. most correlated**

- D. all available

Explanation: Decision trees split data using the most correlated (informative) features at each node — typically chosen by metrics like Gini impurity or information gain. They do not use all features or only numerical ones at every split.

Q5. Which of the following is NOT a characteristic of cloud computing?

- A. On-demand self-service
- B. Broad network access

✓ **C. Fixed resource allocation**

- D. Measured service

Explanation: Cloud computing is defined by on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service (NIST). Fixed resource allocation is the opposite of cloud — clouds are elastic and scalable.

Q6. What does the OSI model stand for?

✓ **A. Open Systems Interconnection**

- B. Operating System Interface
- C. Open Software Integration
- D. Optical Signal Interchange

Explanation: OSI stands for Open Systems Interconnection. It is a conceptual framework developed by ISO that standardizes the communication functions of a telecommunication or computing system into 7 layers.

Q7. Which of the following is a primary key constraint?

A. Allows NULL values

✓ **B. Must be unique for each record**

C. Can be duplicated

D. Is optional

Explanation: A primary key must be unique for every record and cannot be NULL. It uniquely identifies each row in a table. It is mandatory (not optional) and cannot contain duplicate values.

Q8. In Python, what is the output of: `print(type([]))`?

A. `<class 'tuple'>`

B. `<class 'dict'>`

✓ **C. `<class 'list'>`**

D. `<class 'set'>`

Explanation: `[]` is an empty list literal in Python. `type([])` returns `<class 'list'>`. Tuples use `()`, dicts use `{}`, and sets use `set()` or `{values}`.

Q9. Which layer of the OSI model is responsible for end-to-end communication?

A. Network

✓ **B. Transport**

C. Session

D. Data Link

Explanation: The Transport Layer (Layer 4) of OSI is responsible for end-to-end communication, error recovery, and flow control between hosts. TCP and UDP operate here. The Network layer handles routing; the Session layer manages connections.

Q10. What is the default port for HTTP?

A. 443

B. 21

✓ C. 80

D. 25

Explanation: HTTP uses port 80 by default. HTTPS uses 443, FTP uses 21, and SMTP (email) uses 25. These are well-known port assignments defined by IANA.

Q11. Which of the following best describes polymorphism in OOP?

A. A class can have only one method

✓ B. Objects of different classes can be treated as objects of a common superclass

C. A class can inherit from multiple classes

D. An object cannot be modified after creation

Explanation: Polymorphism allows objects of different types to be treated uniformly through a common interface or superclass. For example, a draw() method behaves differently for Circle and Rectangle, but both can be called the same way. Multiple inheritance is a separate concept.

Q12. In SQL, which command is used to remove all rows from a table without deleting the table structure?

A. DELETE

B. DROP

✓ C. TRUNCATE

D. REMOVE

Explanation: TRUNCATE removes all rows from a table quickly and efficiently without logging individual row deletions, while keeping the table structure intact. DELETE can also remove rows but is logged per row and allows a WHERE clause. DROP deletes the entire table.

Q13. What does RAM stand for?

A. Read Access Memory

✓ **B. Random Access Memory**

C. Rapid Application Memory

D. Read And Modify

Explanation: RAM stands for Random Access Memory — volatile memory that allows any memory location to be accessed in the same time regardless of its physical location. It is the primary working memory of a computer.

Q14. Which sorting algorithm has the best average-case time complexity?

A. Bubble Sort

B. Insertion Sort

✓ **C. Merge Sort**

D. Selection Sort

Explanation: Merge Sort has $O(n \log n)$ average-case time complexity, which is optimal for comparison-based sorting. Bubble Sort, Insertion Sort, and Selection Sort all have $O(n^2)$ average-case complexity, making them much slower for large datasets.

Q15. Which class of IP addresses is reserved for experimental use?

A. Class A

✓ **B. Class E**

C. Class C

D. Class D

Explanation: Class E (240.0.0.0–255.255.255.254) is reserved by IANA for experimental purposes and is not used for public or private networking. Class D is for multicast; A, B, C are standard unicast classes.

Q16. Which of the following search algorithms guarantees both completeness and optimality, assuming the heuristic is admissible and consistent?

A. Greedy Best-First Search

✓ **B. A* Search**

C. Depth-First Search

D. Hill Climbing

Explanation: A* Search is both complete (will find a solution if one exists) and optimal (finds the least-cost path) when the heuristic is admissible and consistent. Greedy Best-First is not optimal; DFS and Hill Climbing are neither complete nor optimal in general.

Q17. Which protocol uses port 443 and operates at Layer 7?

A. FTP

✓ **B. HTTPS**

C. SSH

D. SNMP

Explanation: HTTPS (HTTP Secure) uses port 443 and operates at the Application layer (Layer 7) of the OSI model. FTP uses ports 20/21, SSH uses port 22, and SNMP uses ports 161/162.

Q18. Consider the following method setValue which adds a value into an array list. Assuming list is an array of Strings, which might cause setValue to throw an ArrayIndexOutOfBoundsException exception?

A. the type of the parameter position being declared as an int

B. not declaring the method static

C. the return type being declared as void

✓ **D. the value of the first parameter**

Explanation: An ArrayIndexOutOfBoundsException occurs when an index is outside the array bounds. If 'position' (the first parameter) has a value that results in list[position-1] going out of bounds (e.g., position=0 gives index -1, or position > array length), the exception is thrown. The type, return type, or static modifier don't cause this.

Q19. Which Activity lifecycle callback is the first one called after the Activity is launched and is where initial setup like inflating the UI layout typically occurs?

A. onStart()

✓ B. onCreate()

C. onResume()

D. onAttach()

Explanation: onCreate() is the first callback in the Android Activity lifecycle, called when the Activity is first created. It is where UI inflation (setContentView), variable initialization, and other one-time setup occurs. onStart() and onResume() come after; onAttach() belongs to Fragments.

Q20. Which of the following is known as a set of entities of the same type that share same properties, or attributes?

✓ A. Entity set

B. Relation set

C. Tuples

D. Entity Relation model

Explanation: An Entity Set in the ER (Entity-Relationship) model is a collection of similar entities that share the same attributes. Tuples are rows in a relation; a relation set is a set of relationships between entities.

Q21. Which CSS property is used to control the space between the content and the border of an element?

✓ A. padding

B. margin

C. spacing

D. border-spacing

Explanation: padding controls the space between an element's content and its border (inside the border). margin controls space outside the border. 'spacing' is not a CSS property, and border-spacing applies to table cell spacing.

Q22. Which of the following is not a Boolean expression?

A. $10 == 2 * 5$

✓ B. $20 = 5 + 15$

C. $10 ** 2 != 100$

D. $\text{True} != \text{True}$

Explanation: $20 = 5 + 15$ uses a single '=' which is assignment, not comparison — it is not a Boolean expression. The others use comparison operators ($==$, $!=$) which return True or False.

Q23. In the TCP/IP model, which layer corresponds to the Transport layer of OSI?

✓ A. Transport

B. Application

C. Network Access

D. Internet

Explanation: The TCP/IP model has a Transport layer that directly corresponds to the OSI Transport layer (Layer 4). It handles end-to-end communication, reliability, and flow control via TCP and UDP.

Q24. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation. What integrity constraint dictates this?

A. Referencing

✓ B. Referential

C. Specific

D. Primary

Explanation: Referential Integrity ensures that a foreign key value in one table must match a primary key value in another (referenced) table, or be NULL. This prevents orphan records and maintains consistency across related tables.

Q25. What will be the output of the following code? `var x = 5; var y = '5'; console.log(x == y);`

✓ **A. true**

- B. undefined
- C. null
- D. false

Explanation: JavaScript's `==` operator performs type coercion. It converts '5' (string) to 5 (number) before comparing, so `5 == 5` evaluates to true. Use `===` to avoid coercion (which would return false).

Q26. Which of the following data structures uses the LIFO (Last In First Out) principle?

A. Binary Tree

✓ **B. Stack**

- C. Linked List
- D. Queue

Explanation: A Stack follows LIFO: the last element pushed is the first to be popped. A Queue uses FIFO (First In First Out). Binary Trees and Linked Lists do not inherently follow LIFO.

Q27. HMMs are useful for modeling sequential data where the underlying states are ____.

A. Pre-defined

✓ **B. partially observable**

- C. always observable
- D. completely random

Explanation: Hidden Markov Models (HMMs) model systems where the state sequence is hidden (not directly observed) — only the outputs/emissions are visible. This 'partial observability' is what makes them 'hidden.' They are not random or fully observable.

Q28. Mutual execution algorithms are primarily used to resolve:

- A. None
- B. Busy-waiting
- C. Deadlock
- D. Starvation

✓ **E. Race condition**

Explanation: Mutual exclusion algorithms (like mutexes, semaphores) prevent multiple processes from entering a critical section simultaneously, resolving race conditions. Busy-waiting is a method (not what is resolved), and deadlock/starvation are separate problems.

Q29. In COCOMO model, Initial estimate E_i = ____

- A. $a \cdot (KLOC)^b$
- B. $b \cdot (KLOC)^a$

✓ **C. $a(KLOC)^b$**

- D. $b(KLOC)^a$

Explanation: In the COCOMO (Constructive Cost Model), the initial effort estimate is $E_i = a \cdot (KLOC)^b$, where a and b are constants depending on the project type, and KLOC is thousands of lines of code. Option c represents this correctly in notation.

Q30. What will be printed when the following code is executed? `tuple1 = (1, 2, 3, 4); print(tuple1[1:-1])`

- A. (2, 3)
- B. Error: tuple slicing

✓ **C. (2, 3, 4)**

- D. (1, 2, 3, 4)

Explanation: Python tuple slicing `[1:-1]` starts from index 1 (value 2) and goes up to but not including the last element (index -1, value 4). So `(1,2,3,4)[1:-1] = (2,3,4)`. Tuple slicing is perfectly valid in Python.

Q31. A process goes from running-state to blocked-state whenever a running process:

- A. moved to secondary storage because of memory shortage
- B. None
- C. interrupted by a higher priority process
- D. CPU executes context switching

✓ **E. needs to get input/output in the middle of execution**

Explanation: A process moves from Running to Blocked when it requests I/O or another event that it must wait for. Preemption by higher priority causes Running→Ready, not Blocked. Memory shortage causes swapping, not the blocked state.

Q32. Which of the following is not true about functions in Python?

- A. Functions are reusable pieces of program
- B. Functions provide better modularity for your programs

✓ **C. Function definitions can alter flow of execution**

- D. Python allows user defined functions

Explanation: Function definitions themselves do not alter the flow of execution — they only define what happens when called. Function calls alter flow, but the definition statement just registers the function. Options a, b, and d are all true about Python functions.

Q33. Which Android UI component is described as a 'microactivity' that represents a behavior or portion of the UI and must be hosted within an Activity?

- A. View
- B. Adapter

✓ **C. Fragment**

- D. Service

Explanation: A Fragment is often called a 'microactivity' — it represents a reusable portion of the UI with its own lifecycle, and must be embedded within an Activity. Views are basic UI elements; Services run in the background; Adapters bind data to views.

Q34. Which method is used to add an element to the end of an array in JavaScript?

A. pop()

✓ B. push()

C. shift()

D. unshift()

Explanation: push() adds one or more elements to the end of a JavaScript array and returns the new length. pop() removes the last element; shift() removes the first; unshift() adds to the beginning.

Q35. Four conditions are required for a deadlock to occur. Which one of the following is not a condition of deadlock?

A. Circular wait

B. Hold and wait

C. Mutual exclusion

D. No preemption

✓ E. Busy wait

Explanation: The four Coffman conditions for deadlock are: Mutual Exclusion, Hold and Wait, No Preemption, and Circular Wait. Busy waiting is a CPU technique (spin-lock), not a condition for deadlock.

Q36. Which command can display the current ARP cache on a Windows system?

A. ipconfig

B. ping

✓ C. arp -a

D. netstat -r

Explanation: arp -a displays the current ARP (Address Resolution Protocol) cache table on Windows, showing IP-to-MAC address mappings. ipconfig shows network interface config; ping tests connectivity; netstat -r shows routing tables.

Q37. In a graph, what does a cycle represent?

- A. A linear path
- B. A spanning tree
- C. A set of disconnected vertices

✓ **D. A path that starts and ends at the same vertex**

Explanation: A cycle in a graph is a path that begins and ends at the same vertex with no repeated edges. Spanning trees are acyclic by definition; a linear path doesn't return to the start; disconnected vertices form no path at all.

Q38. If an algorithm has a time complexity of $O(n^2)$, what happens to the running time if the input size is doubled?

- A. It triples
- B. It doubles

✓ **C. It quadruples**

- D. It remains the same

Explanation: $O(n^2)$ means time is proportional to n^2 . If n doubles to $2n$, time becomes $(2n)^2 = 4n^2$ — it quadruples. This is why $O(n^2)$ algorithms become very slow for large inputs.

Q39. Which of the following is anisotropy regarding the flow of execution in a while statement?

- A. If the condition is yielding not true, execute the body

✓ **B. If the condition is false, exit the while statement and continue execution**

- C. All
- D. Evaluates the condition yielding only True

Explanation: In a while loop, the condition is evaluated first. If false, the loop exits and execution continues after the loop body. Option a is backwards (false=stop, not execute). Option d is wrong — the condition yields True or False.

Q40. The main purpose of STP (Spanning Tree Protocol) is to:

- A. Prioritize traffic
- B. Monitor SNMP traps

✓ **C. Prevent Layer 2 loops**

- D. Filter traffic by VLAN

Explanation: STP prevents Layer 2 (switching) loops in Ethernet networks by selectively blocking redundant paths, creating a loop-free logical topology. VLANs and traffic prioritization are handled by other protocols (802.1Q, QoS).

Q41. Which one of the following scheduling algorithms is most likely to cause starvation?

- A. Round-Robin (RR)
- B. First-Come, First Served (FCFS)

✓ **C. Short-Job First (SJF)**

- D. None
- E. Priority-Based With Aging

Explanation: SJF (Shortest Job First) can cause starvation: long processes may never execute if a constant stream of shorter jobs keeps arriving. Priority-Based With Aging specifically prevents starvation by increasing a process's priority over time. RR and FCFS are fair by design.

Q42. Bereket needs to send an email with an attachment containing sensitive information. Which of the following is the BEST choice to maintain confidentiality?

- A. Digital signature

✓ **B. Encryption**

- C. Hashing
- D. Data masking

Explanation: Encryption converts data into an unreadable format, ensuring only the intended recipient with the key can access the content — maintaining confidentiality. Digital signatures verify authenticity; hashing verifies integrity (not reversible); data masking replaces data for display purposes.

Q43. Consider the setValue method. Assuming list is declared as an array of Strings, which one of the following might cause setValue to throw an ArrayIndexOutOfBoundsException exception?

- A. not declaring the method static
- B. the type of the parameter position being declared as an int

✓ **C. the value of the first parameter**

- D. the method having more than one parameter
- E. the return type being declared as void

Explanation: The value of the first parameter (position) determines the index list[position-1]. If position is 0, the index is -1 (invalid); if position exceeds array size, it's out of bounds. None of the other options affect array index behavior.

Q44. Which of the following best describes a compiled language?

- A. Code is always faster in compiled languages because they have no errors
- B. Code is executed line-by-line

✓ **C. Code is translated directly into machine code before execution**

- D. Compiled languages cannot be used for low-level programming

Explanation: A compiled language translates the entire source code into machine code (binary) before execution via a compiler. Interpreted languages execute line-by-line. Compiled languages can have errors and are actually commonly used for low-level programming (C, C++).

Q45. Which of the following is a feature of Object-Oriented Programming?

- A. Global variables only

✓ **B. Encapsulation**

- C. No inheritance
- D. Linear execution

Explanation: Encapsulation is a core OOP principle — bundling data and methods together in a class and restricting direct access to internal state. Other key OOP features include inheritance, polymorphism, and abstraction.

Q46. What is the result of $5 \% 2$ in Python?

A. 2.5

B. 2

✓ C. 1

D. 0

Explanation: The % operator gives the remainder of division. $5 \div 2 = 2$ remainder 1, so $5 \% 2 = 1$. It does not return the quotient (2) or the decimal result (2.5).

Q47. Which of the following best describes a foreign key?

A. A key that is unique in its table

✓ B. A key that references the primary key of another table

C. A key that is always NULL

D. A key used only in SELECT statements

Explanation: A foreign key is a column (or set of columns) in one table that references the primary key in another table, creating a relationship between the two. It enforces referential integrity.

Q48. In networking, what does DNS stand for?

A. Data Network Service

✓ B. Domain Name System

C. Dynamic Node Server

D. Distributed Name Service

Explanation: DNS stands for Domain Name System. It translates human-readable domain names (like www.example.com) into IP addresses that computers use to communicate on the internet.

Q49. Which of the following is an example of a non-volatile memory?

A. RAM

B. Cache

✓ C. ROM

D. Register

Explanation: ROM (Read-Only Memory) is non-volatile — it retains data even when power is off. RAM, cache, and registers are all volatile memory (data is lost when power is removed).

Q50. What is the purpose of the 'break' statement in Python?

A. To skip the current iteration

✓ **B. To exit the loop entirely**

C. To restart the loop

D. To pause execution

Explanation: break immediately terminates the loop it is in and transfers control to the statement following the loop. continue skips the current iteration but keeps the loop running. There is no 'pause' in Python loops.

Q51. Which of the following HTTP methods is used to send data to a server to create/update a resource?

A. GET

B. DELETE

✓ **C. POST**

D. HEAD

Explanation: POST sends data to the server to create or update a resource. GET retrieves data; DELETE removes a resource; HEAD retrieves headers only. PUT is used for full updates (idempotent), while POST is for creating new resources.

Q52. In a relational database, what is a view?

A. A physical copy of a table

✓ **B. A virtual table based on a SQL query**

C. An index on a table

D. A stored procedure

Explanation: A view is a virtual table created by a stored SQL query. It does not store data itself but presents data from one or more underlying tables. It simplifies complex queries and provides security by restricting column/row access.

Q53. What does the 'static' keyword mean in Java?

A. The variable changes at runtime

✓ **B. The method/variable belongs to the class, not to instances**

C. The variable cannot be modified

D. The method can only be called once

Explanation: A static member belongs to the class itself rather than any object instance. It is shared across all instances and can be accessed via the class name. It does NOT mean constant — that's the 'final' keyword.

Q54. Which of the following is an example of a graph traversal algorithm?

A. Quick Sort

B. Binary Search

✓ **C. Breadth-First Search**

D. Merge Sort

Explanation: BFS (Breadth-First Search) is a graph traversal algorithm that explores all neighbors at the current depth before moving deeper. DFS is another graph traversal. Quick Sort and Merge Sort are sorting algorithms; Binary Search is for sorted arrays.

Q55. What is the main advantage of using a linked list over an array?

A. Faster access by index

✓ **B. Dynamic size**

C. Less memory usage

D. Simpler syntax

Explanation: Linked lists have dynamic size — nodes can be added or removed without pre-allocating fixed memory. Arrays have faster index access ($O(1)$ vs $O(n)$ for linked list), but require contiguous memory and resizing is expensive.

Q56. Which of the following is NOT a valid IP address?

- A. 192.168.1.1
- B. 255.255.255.0
- ✓ C. 300.168.1.1
- D. 10.0.0.1

Explanation: Each octet of an IPv4 address must be between 0 and 255. 300 exceeds this maximum, making 300.168.1.1 invalid. All other options contain valid octets.

Q57. In machine learning, what does 'overfitting' mean?

- ✓ A. The model performs well on training data but poorly on new data
- B. The model is too simple to learn the data
- C. The model has too few parameters
- D. The model performs equally on all datasets

Explanation: Overfitting occurs when a model learns the training data too well, including noise and outliers, so it fails to generalize to new data. Underfitting is the opposite (model too simple). Overfitted models have low training error but high test error.

Q58. Which of the following is a valid way to declare a variable in JavaScript?

- A. `int x = 5;`
- B. `variable x = 5;`
- ✓ C. `let x = 5;`
- D. `declare x = 5;`

Explanation: In JavaScript, variables are declared with `let`, `const`, or `var`. `'int'` is a Java/C keyword; `'variable'` and `'declare'` are not JavaScript keywords. `let` is the modern block-scoped declaration.

Q59. What is normalization in databases?

A. Adding more tables

✓ **B. Reducing data redundancy and improving data integrity**

C. Increasing query speed

D. Adding primary keys to all tables

Explanation: Normalization is the process of organizing a database to reduce redundancy and improve data integrity by decomposing tables into smaller, related ones following normal forms (1NF, 2NF, 3NF, etc.). It does not directly mean adding tables or keys specifically.

Q60. In Python, which method is used to add an element at the end of a list?

A. add()

B. insert()

C. push()

✓ **D. append()**

Explanation: append() adds a single element to the end of a Python list. insert() adds at a specific index; extend() adds multiple elements. push() exists in JavaScript/Stack, not Python lists. add() is for sets.

Q61. Which of the following protocols is used to assign IP addresses automatically?

A. FTP

✓ **B. DHCP**

C. DNS

D. ARP

Explanation: DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and other network configuration (subnet mask, gateway, DNS) to devices on a network. DNS resolves names; ARP resolves IP to MAC; FTP transfers files.

Q62. What is an abstract class?

A. A class with no methods

✓ **B. A class that cannot be instantiated and may have abstract methods**

C. A class that inherits from multiple classes

D. A class with only private variables

Explanation: An abstract class cannot be instantiated directly and is meant to be subclassed. It may contain abstract methods (without implementation) that subclasses must implement. It can also have concrete methods and fields.

Q63. Which of the following is a lossless compression algorithm?

A. JPEG

B. MP3

✓ **C. ZIP**

D. MPEG

Explanation: ZIP (and algorithms like DEFLATE within it) is lossless — the original data can be perfectly reconstructed after decompression. JPEG and MP3 are lossy compression formats for images and audio respectively, where some data is permanently discarded.

Q64. What is the purpose of an index in a database?

A. To store backup data

✓ **B. To speed up data retrieval operations**

C. To define relationships between tables

D. To encrypt data

Explanation: A database index is a data structure (like a B-tree) that speeds up row lookup and retrieval operations. Without an index, the database must scan all rows (full table scan). Indexes trade storage space for faster query performance.

Q65. In Java, what is the difference between == and .equals() for strings?

A. No difference

✓ **B. == compares references; .equals() compares content**

C. == compares content; .equals() compares references

D. .equals() only works with integers

Explanation: In Java, == checks if two references point to the same object in memory. .equals() compares the actual string content/values. Two different String objects with identical text will have == return false but .equals() return true.

Q66. What is the role of the CPU in a computer?

A. Stores long-term data

✓ **B. Executes program instructions**

C. Manages network connections

D. Renders graphics

Explanation: The CPU (Central Processing Unit) is the brain of the computer — it fetches, decodes, and executes program instructions. Long-term data storage is the HDD/SSD's role; graphics are handled by the GPU; network connections are managed by network interfaces.

Q67. Which of the following best describes recursion?

A. A loop that runs indefinitely

✓ **B. A function that calls itself**

C. A method that overrides another

D. A class that inherits from two classes

Explanation: Recursion is when a function calls itself to solve a smaller instance of the same problem, with a base case to stop. It is not an infinite loop (must terminate), not method overriding, and not multiple inheritance.

Q68. In networking, what is a subnet mask used for?

A. To encrypt data

✓ **B. To identify the network and host portions of an IP address**

C. To assign MAC addresses

D. To filter spam emails

Explanation: A subnet mask divides an IP address into the network portion and the host portion. By ANDing the IP address with the subnet mask, routers can determine which network a device belongs to.

Q69. Which data structure is most appropriate for implementing a queue?

A. Stack

B. Binary Search Tree

✓ **C. Linked List**

D. Array only

Explanation: A Linked List is most appropriate for a queue because it supports $O(1)$ insertion at the tail and $O(1)$ removal from the head. Arrays can also implement queues but circular arrays are needed for efficiency. Stacks implement LIFO, not FIFO.

Q70. What is the difference between TCP and UDP?

A. TCP is faster; UDP is slower

✓ **B. TCP is connection-oriented and reliable; UDP is connectionless and faster**

C. UDP guarantees delivery; TCP does not

D. They are identical protocols

Explanation: TCP (Transmission Control Protocol) establishes a connection, guarantees delivery, ordering, and error checking — at the cost of overhead. UDP (User Datagram Protocol) is connectionless, has no delivery guarantee, but is faster — used for streaming, DNS, gaming.

Q71. Which of the following is an example of supervised learning?

A. K-Means Clustering

✓ **B. Linear Regression**

C. Principal Component Analysis

D. Autoencoders

Explanation: Linear Regression is supervised learning — it learns from labeled training data (input-output pairs) to predict continuous values. K-Means, PCA, and Autoencoders are all unsupervised learning methods that work without labeled data.

Q72. In Python, what does the 'pass' statement do?

A. Skips the current function call

B. Terminates the program

✓ **C. Does nothing; acts as a placeholder**

D. Passes a value to a function

Explanation: pass is a null operation in Python — it does nothing and is used as a placeholder where code is syntactically required but no action is needed (e.g., an empty function or class body). It does not skip, terminate, or pass values.

Q73. What is the purpose of version control systems like Git?

A. To speed up code execution

✓ **B. To track changes in code and collaborate with others**

C. To compile code automatically

D. To test code for bugs

Explanation: Version control systems like Git track changes to files over time, allowing you to revert to previous versions, branch for new features, and collaborate with teams. They do not compile, test, or speed up code execution.

Q103. Which of the following function declarations is correct to define a function that accepts a variable number of arguments?

✓ **A. `def func(*args):`**

B. Not possible in python

C. `def func(args*):`

D. `def func(**args):`

Explanation: In Python, `*args` in a function definition allows it to accept a variable number of positional arguments, collected as a tuple. `**kwargs` captures keyword arguments. The syntax `'args*'` is invalid. Python absolutely supports variable argument functions.

Q105. Which algorithm is most appropriate when the cost of actions varies and finding the least-cost solution is important?

A. Depth-First Search

B. Breadth-First Search

C. Greedy Best-First Search

✓ **D. Uniform Cost Search**

Explanation: Uniform Cost Search (UCS) expands the node with the lowest cumulative path cost, guaranteeing the optimal (least-cost) solution when action costs vary. DFS and BFS don't account for varying costs; Greedy is heuristic-based and not guaranteed optimal.

Q106. How can you apply a CSS Grid layout to a container?

A. `display: block;`

✓ **B. `display: grid;`**

C. `display: inline-grid;`

D. `display: flex;`

Explanation: `display: grid;` makes an element a block-level grid container, enabling CSS Grid layout on its children. `display: inline-grid;` also enables grid but makes the container inline. `display: flex;` uses Flexbox instead; `display: block;` is standard block layout.